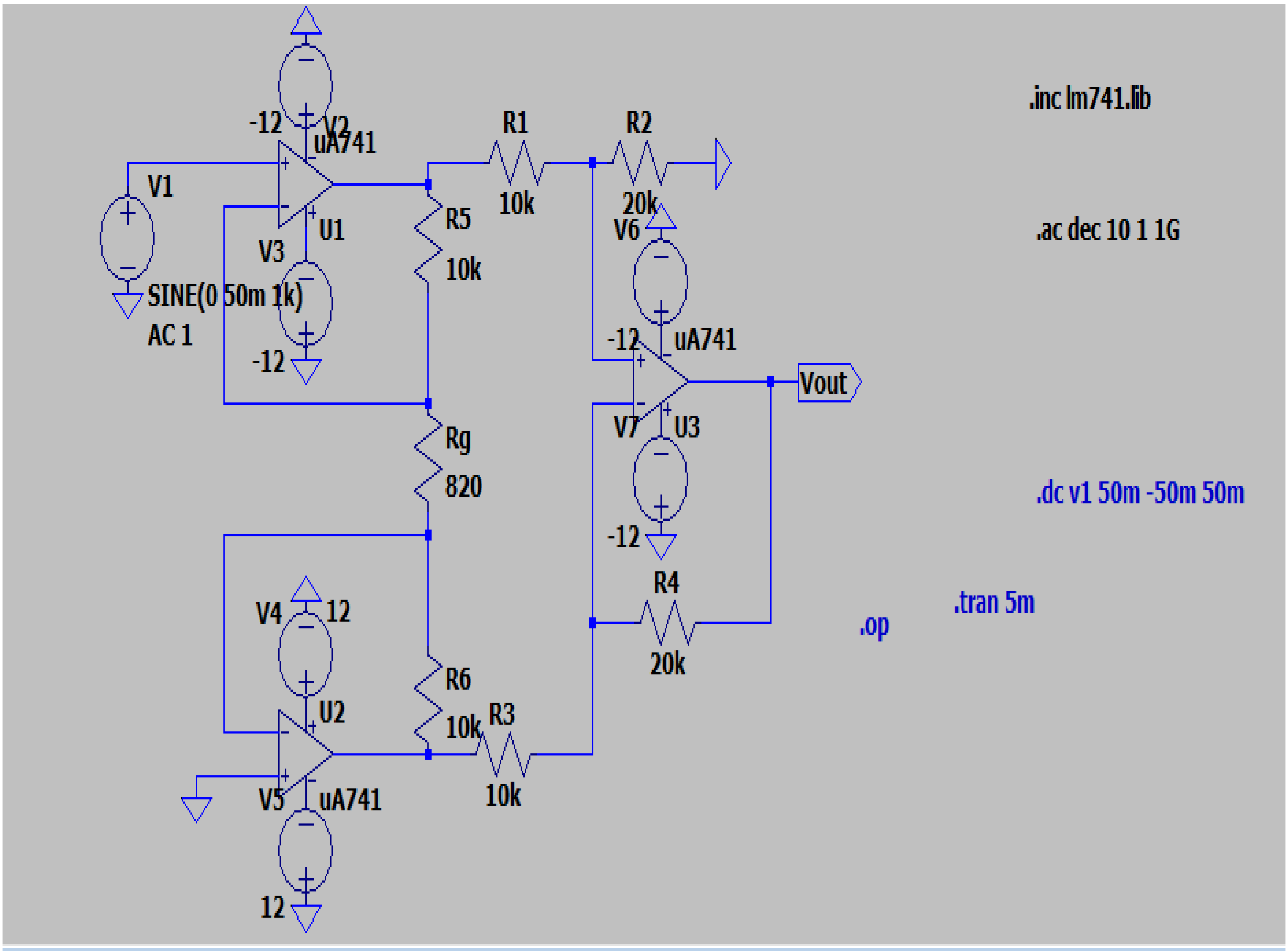
Lab sheet 8- Instrumentation Amplifier

The primary objective is to analyze how varying the gain resistance, Rg, influences the circuit's overall differential gain and to evaluate its effectiveness in rejecting common-mode signals.
By leveraging the high input impedance and superior **Common-Mode Rejection Ratio (CMRR)** characteristic of this configuration, the project explores the practical implementation of high-accuracy signal conditioning.

CIRCUIT DESIGN:



This circuit is designed to provide high precision differential amplification. It consists of two distinct stages:

- **Input Stage (Buffers/Pre-amplifier):** Comprised of op-amps **U1** and **U2**. This stage provides extremely high input impedance, ensuring that the circuit does not load the sensors or signal sources connected to it.
- **Output Stage (Difference Amplifier):** Comprised of op-amp **U3**. This stage subtracts the two signals from the first stage and provides the final gain while rejecting common-mode noise.

Design Parameters & Component Values

Based on your schematic, the following values are used for the gain calculation:

Component	Value	Role
R1, R3	10kohm	Input resistors for the difference amplifier stage
R2, R4	20 kohm	Feedback resistors for the difference amplifier stage
R5, R6	10 kohm	Feedback resistors for the input buffer stage
Rg	820 ohm	**Gain Resistance** (The variable that controls total gain)
V1	5mV(sine)	Differential input signal (1kHz frequency)
Supply	(+12V, -12V)	Dual DC power supply for the uA741 ICs

Theoretical Gain Analysis

The total gain (Av) of an instrumentation amplifier is the product of the gains of the two stages.

First Stage Gain (A1):

The gain of the input stage is determined by R5, R6, and Rg:

$$A_1 = \left(1 + \frac{R_5 + R_6}{R_g}\right)$$

Substituting your values:

$$A_1 = (1 + \frac{10k+10k}{820}) = (1 + \frac{20000}{820}) \approx \mathbf{25.39}$$

Second Stage Gain (A2):
The gain of the difference amplifier stage is the ratio of the feedback to input resistors:

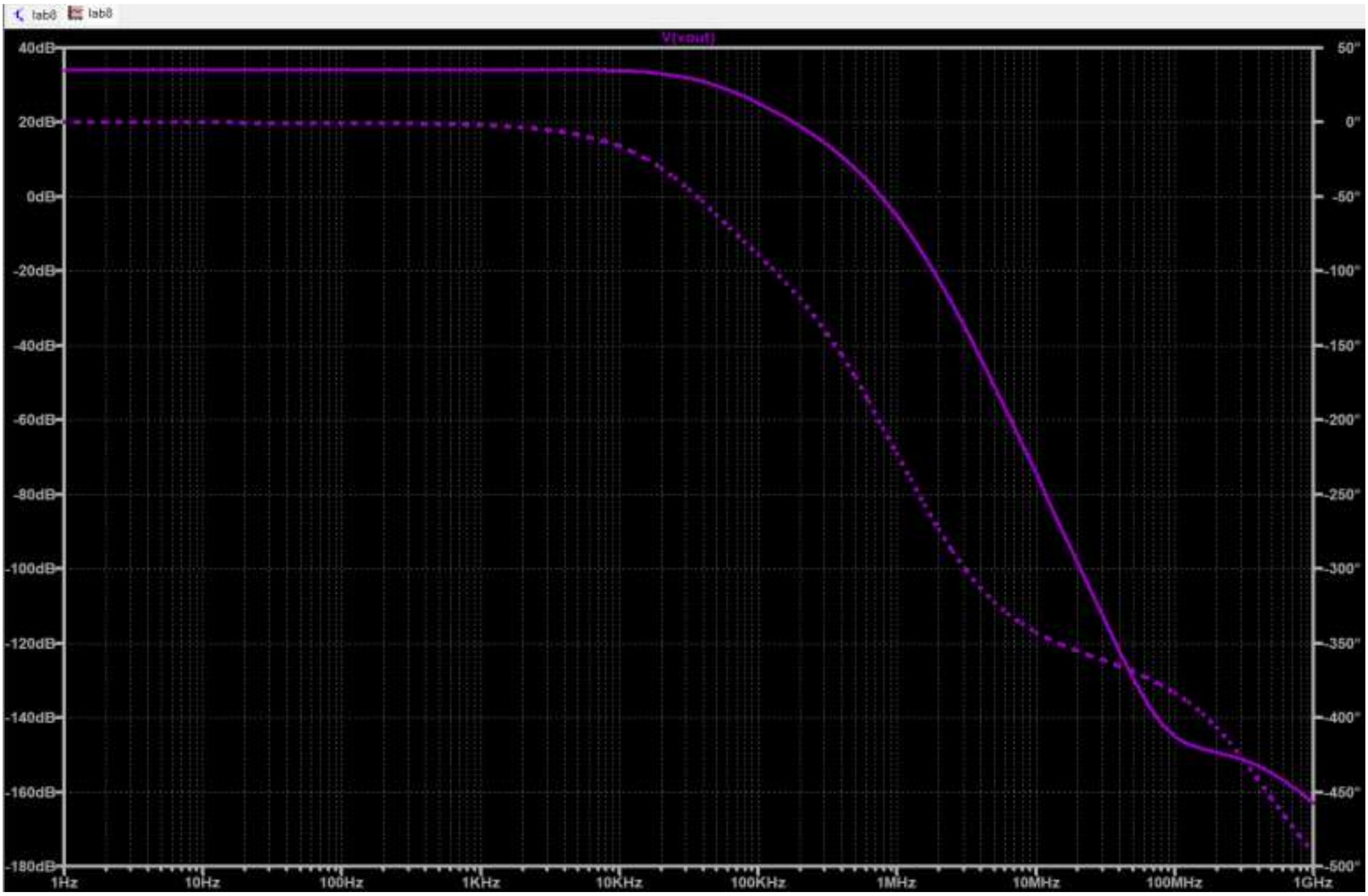
$$A_2 = \frac{R_2}{R_1} = \frac{20k}{10k} = \mathbf{2}$$

Total System Gain (Atotal):

$$A_{total} = A_1 \times A_2 = 25.39 \times 2 \approx \mathbf{50.78}$$

For your 50 mV input, you should expect an output voltage (V out) of approximately 2.54 V.

<u>AC ANALYSIS:</u>



The AC analysis of the instrumentation amplifier reveals the circuit's performance across a wide range of frequencies, specifically focusing on the Gain (Magnitude) and Phase response

<u>Magnitude Response (Solid Line) :</u>

--<u>Mid-band Gain</u>: The simulation shows a constant flat gain at low to medium frequencies. The magnitude is approximately 34.1 dB.

+1

--<u>Conversion Check</u>: A gain of 34.1 dB corresponds to a linear voltage gain of approximately 10 34.1/20 ≈50.7 V/V, which perfectly aligns with the theoretical calculation based on the resistor values (R g =820 Ω).

+2

--<u>Roll-off and Bandwidth</u>: The gain remains stable until approximately 40 kHz, after which it begins to decrease at a rate of −20 dB/decade (for a single pole) or more as it encounters the internal compensation poles of the μA741 op-

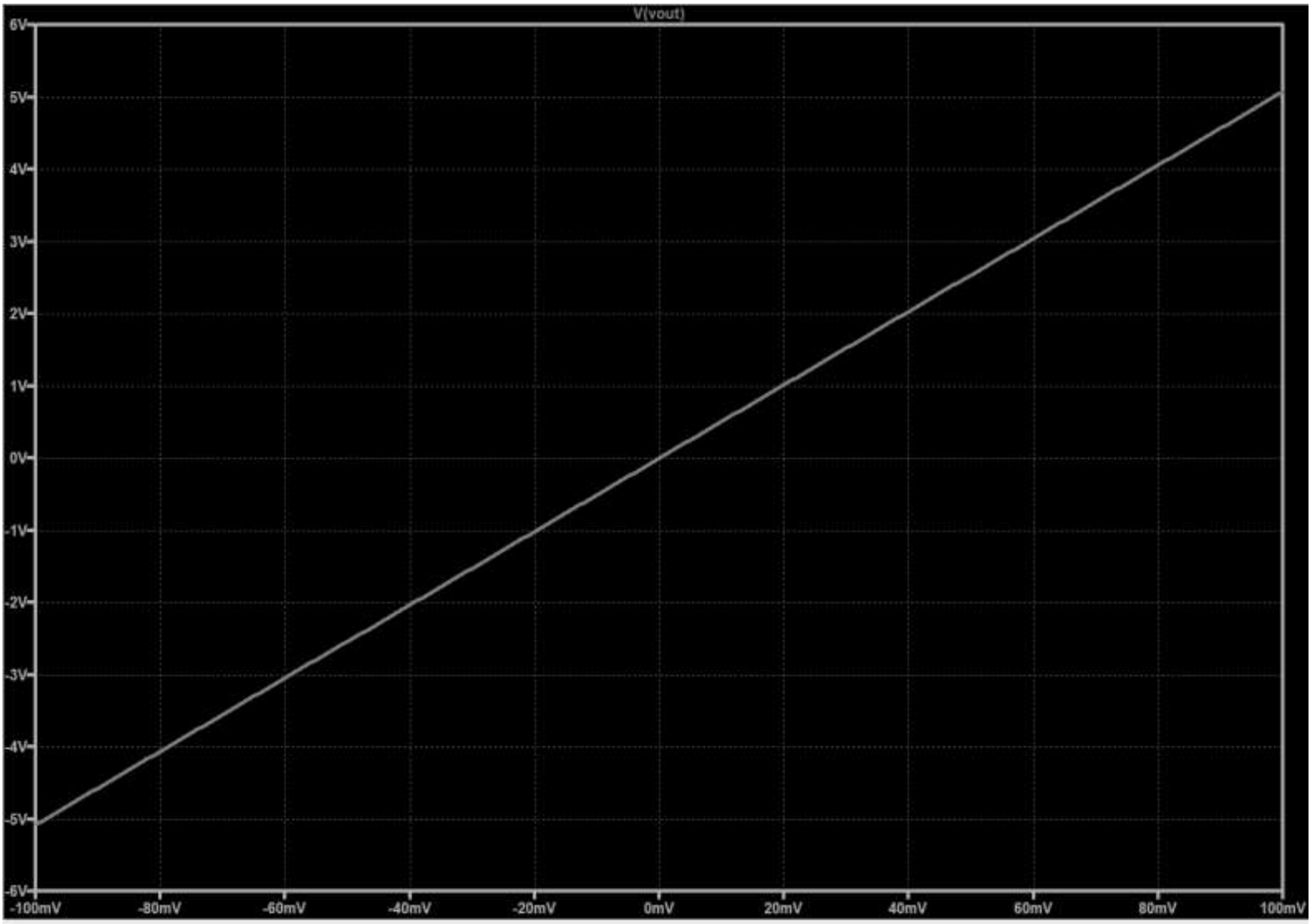
amps.
+1
--<u>Unity Gain Bandwidth</u>: The gain crosses the 0 dB line (where V out =V in) at roughly 1 MHz, which is the typical Gain-Bandwidth Product (GBP) for a standard 741 IC.

Phase Response (Dotted Line)
--<u>Phase Shift</u>: At low frequencies, the phase starts at 0 ∘ , indicating that the output is in phase with the differential input.
+1
--<u>Phase Lag:</u> As the frequency increases towards the corner frequency, a phase lag is introduced. At the frequency where the gain drops by 3 dB, the phase shift is approximately −45 ∘ .
--<u>High-Frequency Behavior</u>: At extremely high frequencies (above 10 MHz), the phase shift reaches significant values (beyond −180 ∘), which is characteristic of the multi-stage internal architecture of the op-amps used.

Application Suitability: The flat gain response in the 10 Hz to 10 kHz range makes this instrumentation amplifier ideal for low-frequency applications such as patient monitoring (ECG/EEG) or industrial sensor signal conditioning, where the signals of interest fall within this bandwidth.
+1
Stability: The smooth roll-off in magnitude and predictable phase transition suggest that the amplifier remains stable within its intended operating frequency range.

Parameter	Observed Value
Max Magnitude (dB)	≈34.1 dB
Linear gain(Av)	≈50.7 V/V
3-dB Bandwidth	≈40 kHz
Phase at Mid-band	0 ∘

<u>DC SWEEP ANALYSIS AND LINEARITY:</u>



The DC sweep plot shows the transfer characteristic of the instrumentation amplifier, specifically the relationship between the differential input voltage (V in) and the output voltage (V out).

1. **Linearity and Dynamic Range Linear Behavior:** The plot is a perfectly straight line, which indicates that the amplifier has excellent linearity. This means the gain remains constant across the entire input range of -100 mV to +100 mV.

No Saturation: The output ranges from approximately -5 V to +5 V. Since your supply voltage is ±12V, the op-amps are operating well within their linear region and are not clipping or saturating.

2. **Slope (Gain) Verification** We can verify the gain by picking two points on the line:

At V in =0 mV: The output V out is 0 V. This confirms that the amplifier has minimal DC offset.

At V in =100 mV: The output V out is approximately 5 V.
Calculation: Gain= ΔV in
ΔV out

100 mV 5 V =50 V/V.
This matches our theoretical calculation of 50.78 almost perfectly.

DC OP POINT:

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       --- Operating Point ---

V(n001):        0               voltage
V(n002):        0.000816896     voltage
V(n003):        1.30895e-05     voltage
V(n004):        1.9248e-05      voltage
V(n005):        12              voltage
V(n006):        3.2341e-05      voltage
V(n007):        1.9248e-05      voltage
V(n008):        0.000816896     voltage
V(p001):        -12             voltage
V(p002):        12              voltage
V(p003):        -12             voltage
V(p004):        12              voltage
V(p005):        -12             voltage
V(vout):        5.85271e-05     voltage
I(R1):          -8.03807e-08    device_current
I(R2):          -6.54475e-10    device_current
I(R3):          -7.84555e-08    device_current
I(R4):          1.30931e-09     device_current
I(R5):          7.97648e-08     device_current
I(R6):          -7.97648e-08    device_current
I(Rg):          -1.78732e-15    device_current
I(V1):          -7.97262e-08    device_current
I(V2):          0.00133675      device_current
I(V3):          0.00133654      device_current
I(V4):          -0.00133654     device_current
I(V5):          -0.00133675     device_current
I(V6):          0.00133675      device_current
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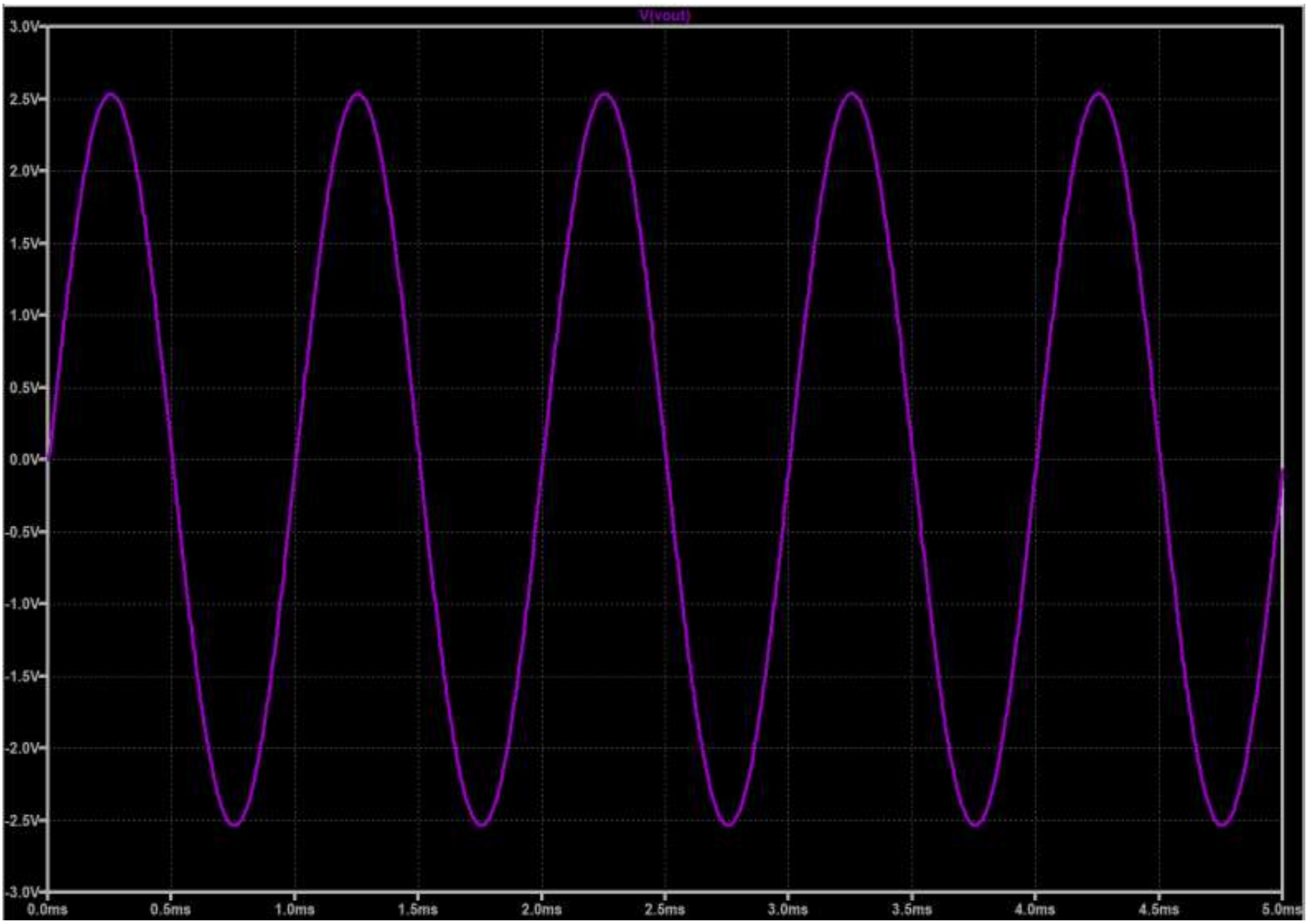
I(V7):          0.00133654    device_current
Ix(u1:1):       7.97262e-08   subckt_current
Ix(u1:2):       7.97648e-08   subckt_current
Ix(u1:3):       0.00133654    subckt_current
Ix(u1:4):       -0.00133675   subckt_current
Ix(u1:5):       -1.60144e-07  subckt_current
Ix(u2:1):       7.97262e-08   subckt_current
Ix(u2:2):       7.97648e-08   subckt_current
Ix(u2:3):       0.00133654    subckt_current
Ix(u2:4):       -0.00133675   subckt_current
Ix(u2:5):       -1.58219e-07  subckt_current
Ix(u3:1):       7.97262e-08   subckt_current
Ix(u3:2):       7.97648e-08   subckt_current
Ix(u3:3):       0.00133654    subckt_current
Ix(u3:4):       -0.00133675   subckt_current
Ix(u3:5):       -1.30787e-09  subckt_current
```

1. **Node Voltage Summary**

--Supply Rails: The results confirm stable power delivery with V(p002),V(p004)=12 V and V(p001),V(p003),V(p005)=−12 V, matching the intended ±12 V dual supply.
+1
--Output Offset (V out): The quiescent output voltage is 5.85 µV (5.85271e−05).
Inference: This extremely low value indicates that the circuit is well-balanced and the µA741 models have very little input offset error in this configuration.
Input Nodes: V(n001) is exactly 0 V, as it is grounded in your schematic for this specific test.

TRANSIENT ANALYSIS:



The graph displays the output voltage (V out) over a 5 ms duration, providing a visual confirmation of the amplifier's gain and signal integrity.

1. **Waveform Characteristics Signal Type**: The output is a clean, undistorted sine wave, which confirms that the amplifier is operating linearly without any clipping or harmonic distortion.

Amplitude: The peak output voltage (V p) is approximately 2.54 V.
Given your input was a 50 mV peak sine wave, the practical gain is:
A v
V in(peak)
V out(peak)

0.05 V 2.54 V =50.8 V/V This perfectly matches your theoretical design of 50.78 V/V.
Frequency: There are exactly 5 full cycles within the 5 ms window.

T= 5 cycles 5 ms =1 ms per cycle.
f= T 1 =1 kHz.
This confirms the output frequency matches the input source V_1 without any phase-locked loop issues or frequency shifting.

2. **Instrumentation Amplifier Performance Metrics Phase Alignment:** The output starts at 0 V and moves positive, indicating the amplifier is non-inverting relative to the differential input ($V_1 - V_2$), as expected from the standard three op-amp configuration.

Voltage Swing: The signal swings symmetrically between +2.54 V and −2.54 V. This symmetry is a result of the dual ±12 V supply and precise resistor matching in the difference amplifier stage.